

Scalability & Future Plan

Date	15 July 2024
Team ID	SWTID-2026-3798
Project Name	Rising Waters-Flood Prediction System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.N o	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Model trained on limited historical data from specific regions	May not generalize well to regions with different climate patterns	High
2	Application deployed only locally, not yet on cloud	Limited accessibility; can't be used by multiple teams simultaneously	High
3	No real-time weather API integration (manual data entry required)	Slower response time during actual emergencies	Medium

Scalability Plan:

S.N o	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-user local Flask app, handles one request at a time	Deploy on IBM Cloud with load balancing to support multiple concurrent users
2	Data Storage	Static CSV dataset stored locally	Migrate to a cloud database (e.g., IBM Cloudant or PostgreSQL) for scalable, real-time data storage
3	Performance	Model inference runs on local machine resources	Use cloud-based GPU/CPU scaling and caching to reduce prediction latency under high load
4	Security	No authentication or data encryption; open local access only	Add user authentication, HTTPS encryption, and secure API endpoints before public deployment

Future Roadmap:

Phase	Planned Feature / Enhancement Target Timeline Expected Impact
Phase 1	Planned Feature/Enhancement:Deploy the Flask application on IBM Cloud Target Timeline:1 month Expected Impact:Makes the system accessible remotely to authorities and disaster teams ,not just locally
Phase 2	Planned Feature/Enhancement:Integrate real-time weather API for automatic data fetching Target Timeline:2 months Expected Impact:Reduces manual data entry and speeds up flood risk predictions during emergencies
Phase 3	Planned Feature/Enhancement:Add user authentication and role-based access(admin,analyst,public) Target Timeline:3 months Expected Impact:Improves security and allow controlled access for different stakeholder types
Phase 4	Planned Feature/Enhancement:Expand model to support multi-region and multi-language support Target Timeline:4-6 Expected Impact:Increases usability across broader geographic areas and diverse user bases